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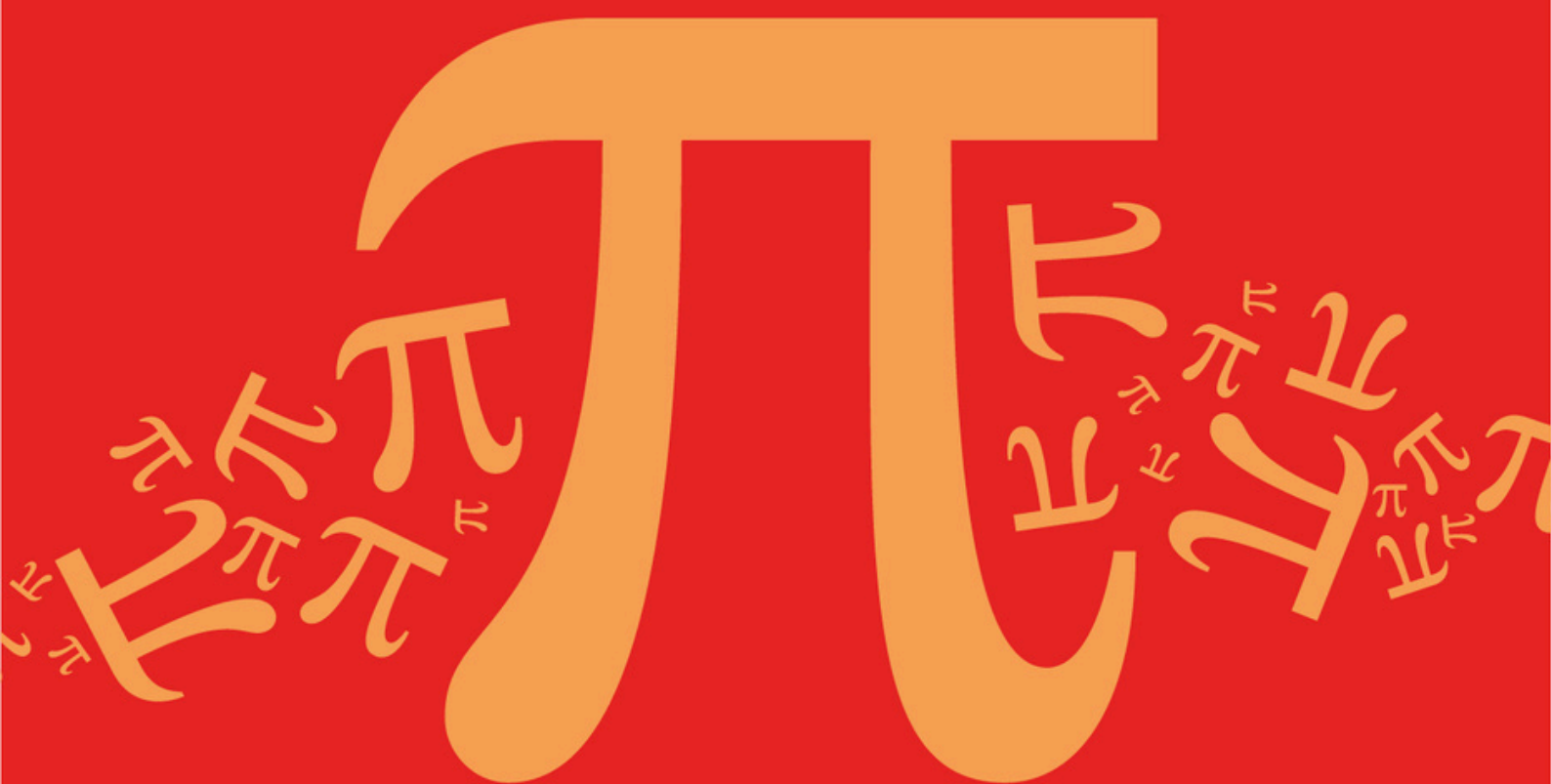
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AQA A Level Further Mathematics (7367)



Topic

DA: Graphs

Sub topic

DA6: Simple Graphs and Trees

Topic Questions

AQA A Level Further Mathematics (7367)

Question Paper

Topic

DA: Graphs

Sub topic

DA6: Simple Graphs and Trees

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Name : _____

Date : _____

Time : _____

Total Marks Available : _____

Total Marks Received : _____

Q1.

A linear programming problem has the constraints

$$1 \leq x \leq 6$$

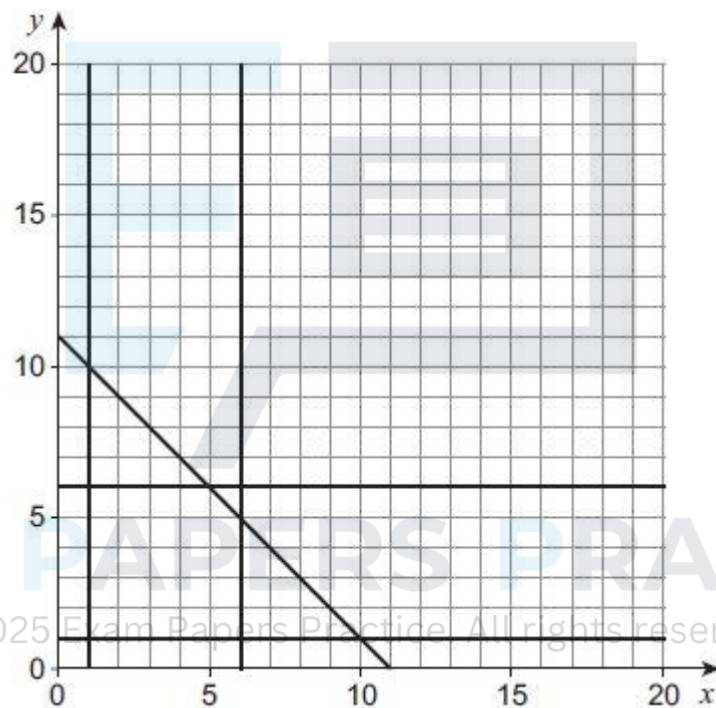
$$1 \leq y \leq 6$$

$$y \geq x$$

$$x + y \leq 11$$

- (a) (i) Complete **Figure 1** to identify the feasible region for the problem.

Figure 1



(2)

- (ii) Determine the maximum value of $5x + 4y$ subject to the constraints.

(2)

- (b) The simple-connected graph G has seven vertices.

The vertices of G have degree 1, 2, 3, v , w , x and y

- (i) Explain why $x \geq 1$ and $y \geq 1$

(1)

- (ii) Explain why $x \leq 6$ and $y \leq 6$

(1)

- (iii) Explain why $x + y \leq 11$.

(2)

- (iv) State an additional constraint that applies to the values of x and y in this context.

(1)

(c) The graph G also has eight edges. The inequalities used in part (a)(i) apply to the graph G .

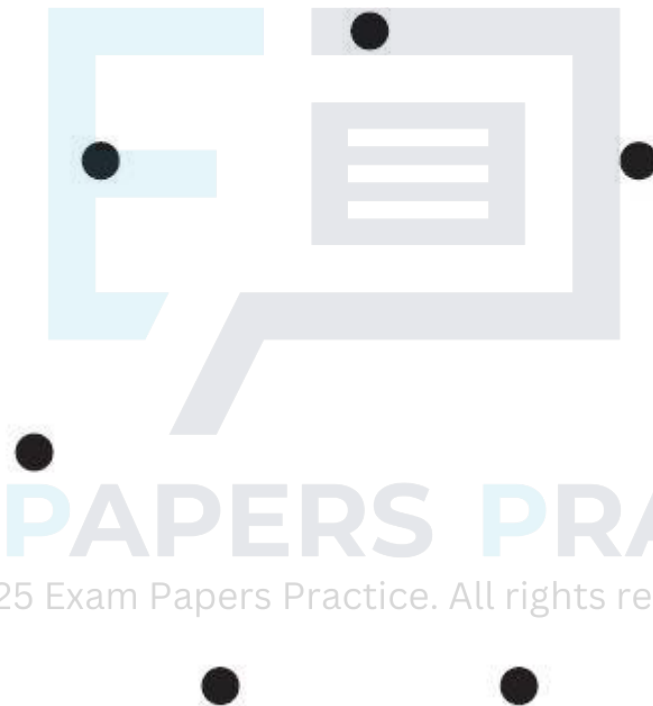
(i) Given that $v + w = 4$, find all the feasible values of x and y .

(3)

(ii) It is also given that the graph G is semi-Eulerian.

On **Figure 2**, draw G .

Figure 2



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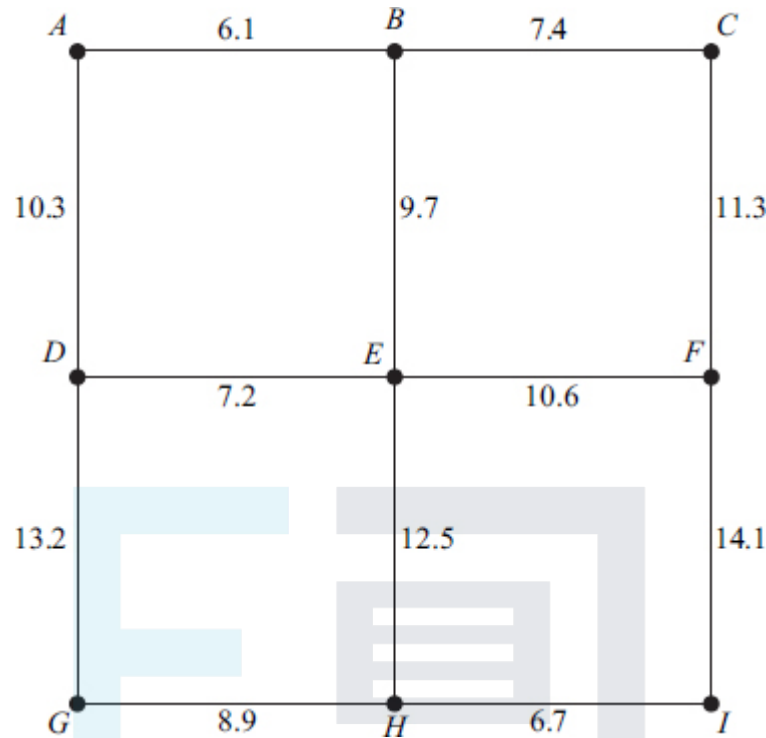
(2)

(Total 14 marks)

Q2.

The following network shows the lengths, in miles, of roads connecting nine villages, $A, B, \dots, I$.

A programme of resurfacing some roads is undertaken to ensure that each village can access all other villages along a resurfaced road, while keeping the amount of road to be resurfaced to a minimum.



- (a) (i) Use Prim's algorithm starting from A , showing the order in which you select the edges, to find a minimum spanning tree for the network.

(ii) State the length of your minimum spanning tree.

(iii) Draw your minimum spanning tree.

(7)

- (b) Given that Prim's algorithm is used with different start vertices, state the final edge to be added to the minimum spanning tree if:

(i) the start vertex is E ;

(ii) the start vertex is G .

(2)

- (c) Given that Kruskal's algorithm is used to find the minimum spanning tree, state which edge would be:

(i) the first to be included in the tree;

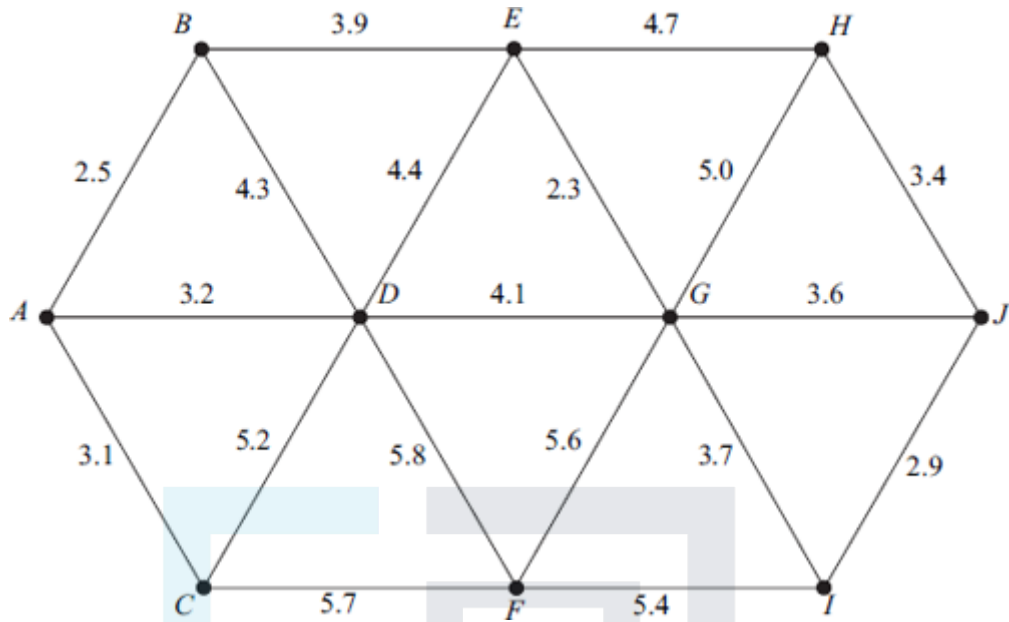
(ii) the last to be included in the tree.

(2)

(Total 11 marks)

Q3.

The following network shows the lengths, in miles, of roads connecting ten villages, $A, B, C, \dots, J$.



- (a) (i) Use Kruskal's algorithm, showing the order in which you select the edges, to find a minimum spanning tree for the network.
- (ii) Find the length of your minimum spanning tree.
- (iii) Draw your minimum spanning tree.

(7)

- (b) Prim's algorithm from different starting points produces the same minimum spanning tree. State the final edge that would be added to complete the minimum spanning tree if the starting point were:

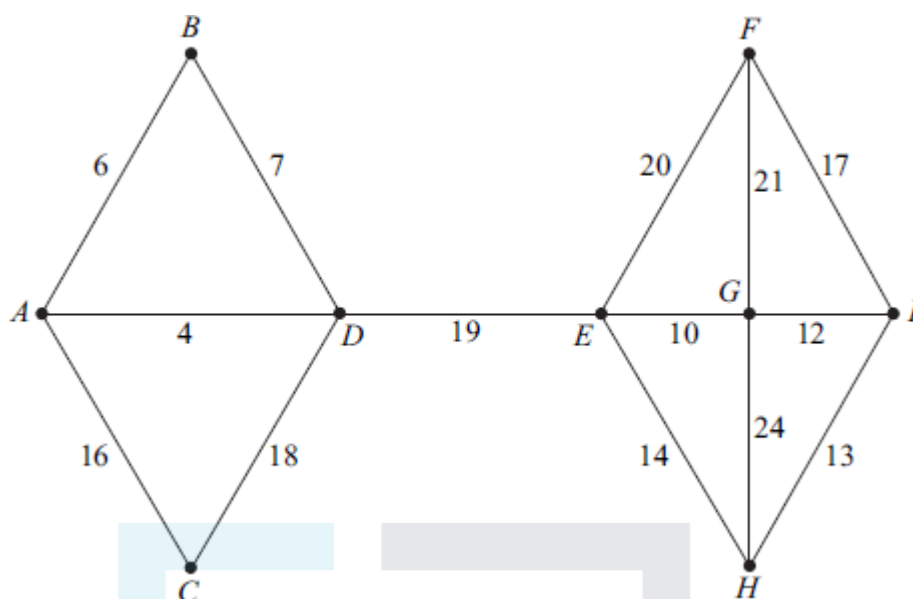
- (i) A;
- (ii) F.

(2)

(Total 9 marks)

Q4.

The following network shows the lengths, in miles, of roads connecting nine villages, A, B, ..., I.



- (a) (i) Use Prim's algorithm starting from A , showing the order in which you select the edges, to find a minimum spanning tree for the network. (4)
- (ii) State the length of your minimum spanning tree. (1)
- (iii) Draw your minimum spanning tree. (2)
- (b) Prim's algorithm from different starting points produces the same minimum spanning tree for this network. State the final edge that would complete the minimum spanning tree using Prim's algorithm:
- (i) starting from D ; (1)
- (ii) starting from H . (1)

(Total 9 marks)

Q5.

A simple connected graph has six vertices.

- (a) One vertex has degree x . State the greatest and least possible values of x . (2)
- (b) The six vertices have degrees

$$x - 2, x - 2, x, 2x - 4, 2x - 4, 4x - 12$$

Find the value of x , justifying your answer.

(2)
(Total 4 marks)

Q6.

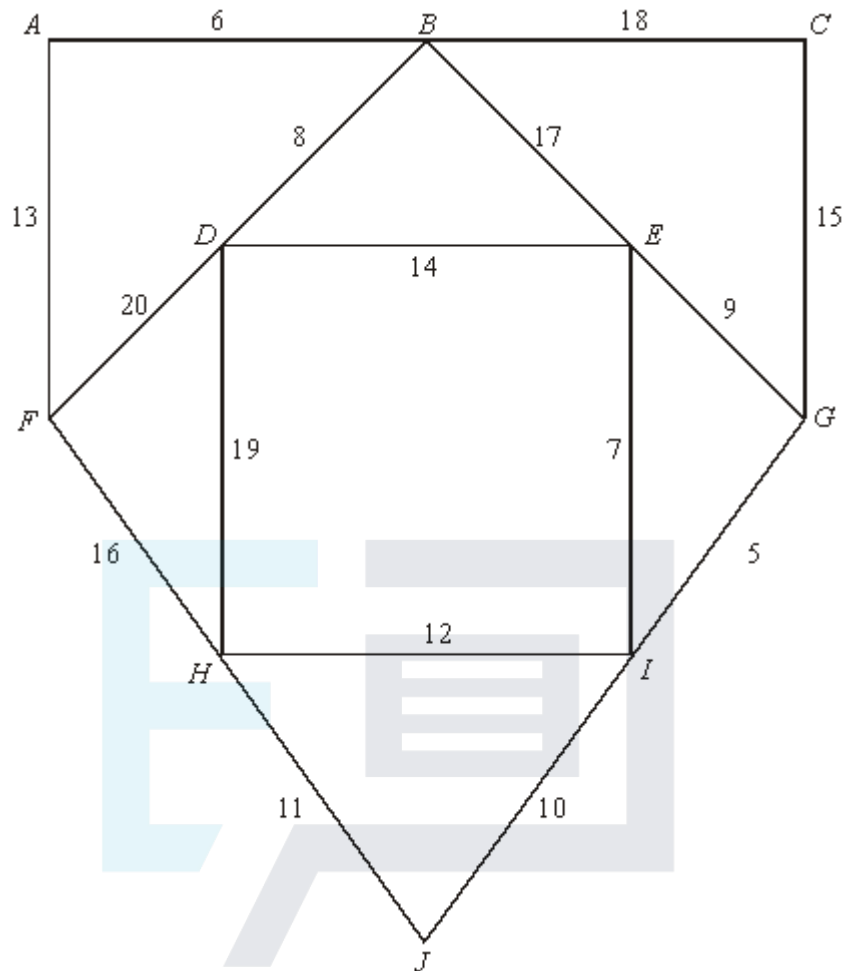
A connected graph G has five vertices and has eight edges with lengths 8, 10, 10, 11, 13, 17, 17 and 18.

- (a) Find the minimum length of a minimum spanning tree for G .(2)
 - (b) Find the maximum length of a minimum spanning tree for G .(2)
 - (c) Draw a sketch to show a possible graph G when the length of the minimum spanning tree is 53.(3)
- (Total 7 marks)**

Q7.

- (a)
 - (i) State the number of edges in a minimum spanning tree of a network with 10 vertices.(1)
 - (ii) State the number of edges in a minimum spanning tree of a network with n vertices.(1)
- (b) The following network has 10 vertices: $A, B, \dots, J$. The numbers on each edge represent the distances, in miles, between pairs of vertices.

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- (i) Use Kruskal's algorithm to find the minimum spanning tree for the network. (5)
- (ii) State the length of your spanning tree. (1)
- (iii) Draw your spanning tree. (2)
- (Total 10 marks)

Q8.

A connected graph has five vertices and has arc lengths of

4, 7, 7, 7, 8, 8, 9 and 12 units.

- (a) State the minimum length of a minimum spanning tree for any such graph. (1)
- (b) State the minimum length of a Hamiltonian cycle for any such graph. (1)
- (c) State the minimum length of an Eulerian cycle for any such graph. (1)
- (d) In the case when the length of its minimum spanning tree is 26 units, draw a sketch

to show a possible graph.

(3)

(Total 6 marks)

Q9.

A graph G has five vertices.

- (a) (i) Given that G is connected, state the number of arcs on a minimum spanning tree.

(1)

- (ii) Given that G is Eulerian with all vertices having the same degree d , state the minimum value of d .

(1)

- (iii) Given that G has a Hamiltonian cycle, state the number of edges in such a cycle.

(1)

- (b) Given that G has all the properties mentioned in part (a), draw a possible graph G .

(1)

(Total 4 marks)

Q10.

The following network has 16 vertices.

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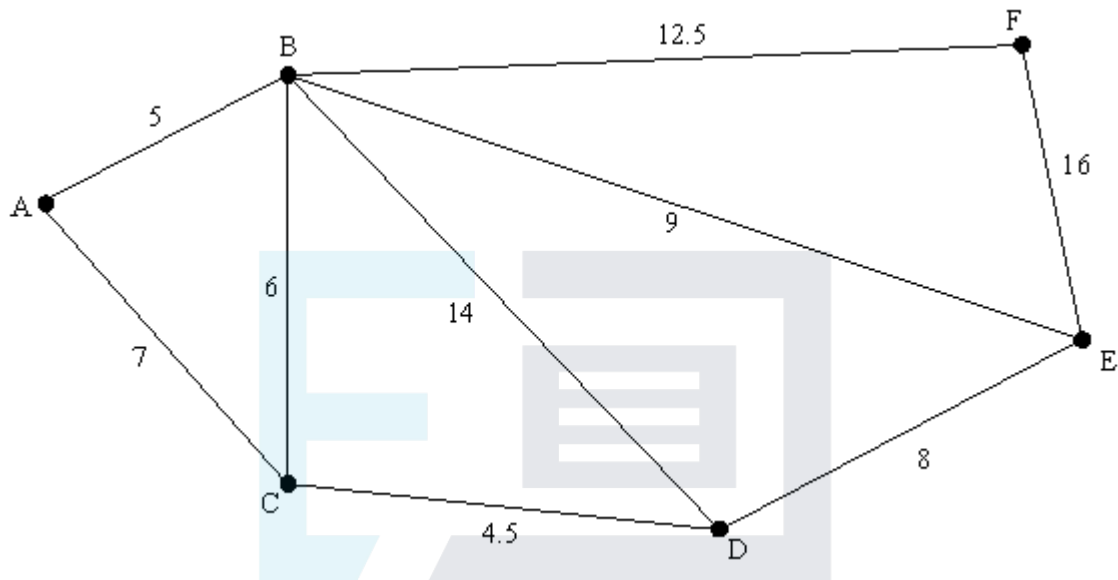
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(2)
(Total 10 marks)

Q11.

A local council is responsible for gritting roads.

- (a) The following diagram shows the lengths of roads, in miles, that have to be gritted.



The gritter is based at A and must travel along all the roads, at least once, before returning to A .

- (i) Explain why it is **not** possible to start from A and, by travelling along each road only once, return to A .

(1)

- (ii) Find an optimal 'Chinese postman' route around the network, starting and finishing at A . State the length of your route.

(6)

- (b) (i) The connected graph of the roads in the area run by another council has six odd vertices. Find the number of ways of pairing these odd vertices.

(1)

- (ii) For a connected graph with n odd vertices, find an expression for the number of ways of pairing these vertices.

(2)

(Total 10 marks)

Q12.

- (a) Draw a simple connected graph with 9 vertices all having degree 2.

(2)

- (b) A simple connected graph has 9 vertices, all having the same degree d .

- (i) State the possible values of d .

(2)

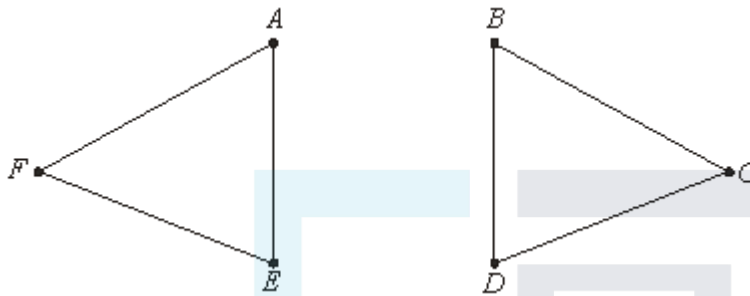
- (ii) For **each** of these values of d , state the number of edges of the graph.

(3)

(Total 7 marks)

Q13.

The graph **G** has six vertices $A - F$ and is illustrated below.



- (a) What is the minimum number of edges which must be added to **G** in order to make the graph connected?

(1)

- (b) What is the minimum number of edges which must be added to **G** in order to make the graph Hamiltonian? State a suitable set of edges.

(2)

- (c) (i) State the minimum number of edges which must be added to **G** in order to create a semi-Eulerian graph. Draw such a graph and write down a trail of this graph which uses all its edges.

(3)

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- (ii) Show that by adding two edges between the same pair of vertices you can create an Eulerian graph.

(1)

- (iii) State the minimum number of edges which must be added to **G** in order to create a **simple** graph which is Eulerian. Draw such a graph.

(3)

(Total 10 marks)

Q14.

- (a) A simple graph has five vertices and their degrees are d , $d + 1$, $d + 1$, $d + 2$ and $d + 3$.

- (i) By considering the sum of the degrees, show that d is odd.

(2)

- (ii) Deduce that $d = 1$ and draw a graph with vertices having the given degrees.

(3)

- (b) A simple graph has 10 vertices.

- (i) Explain why the degree of each vertex must be in the set

$\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

(1)

- (ii) Show that the degrees of the vertices cannot all be different.

(2)

(Total 8 marks)



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